Chapter-4

Video Sequence Processing

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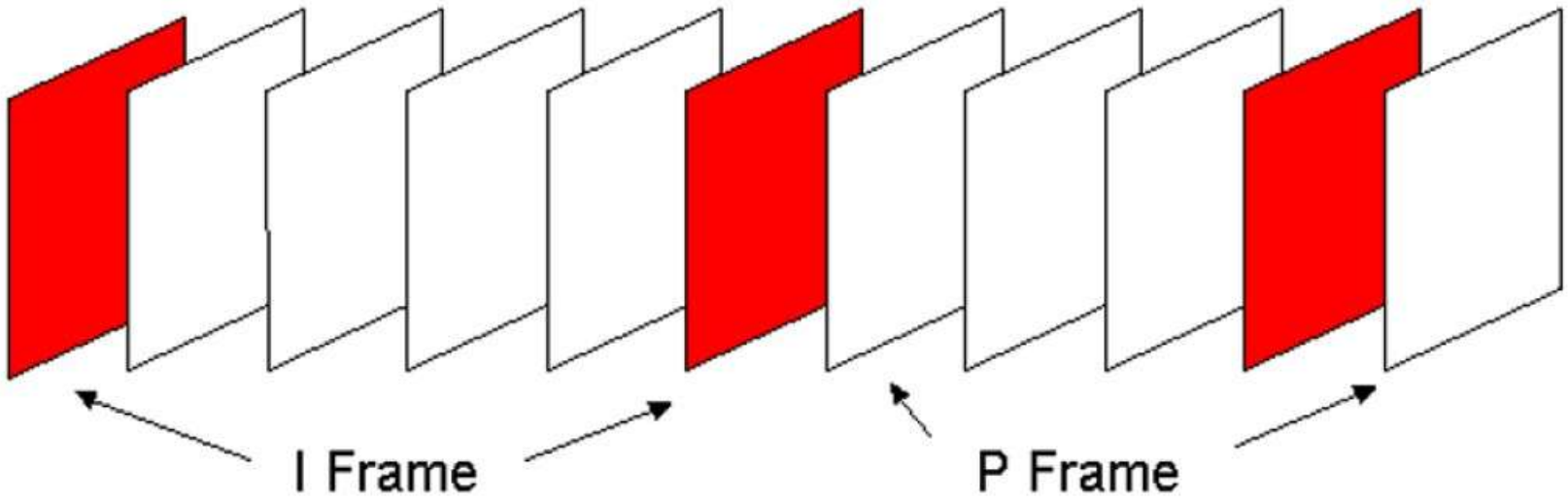
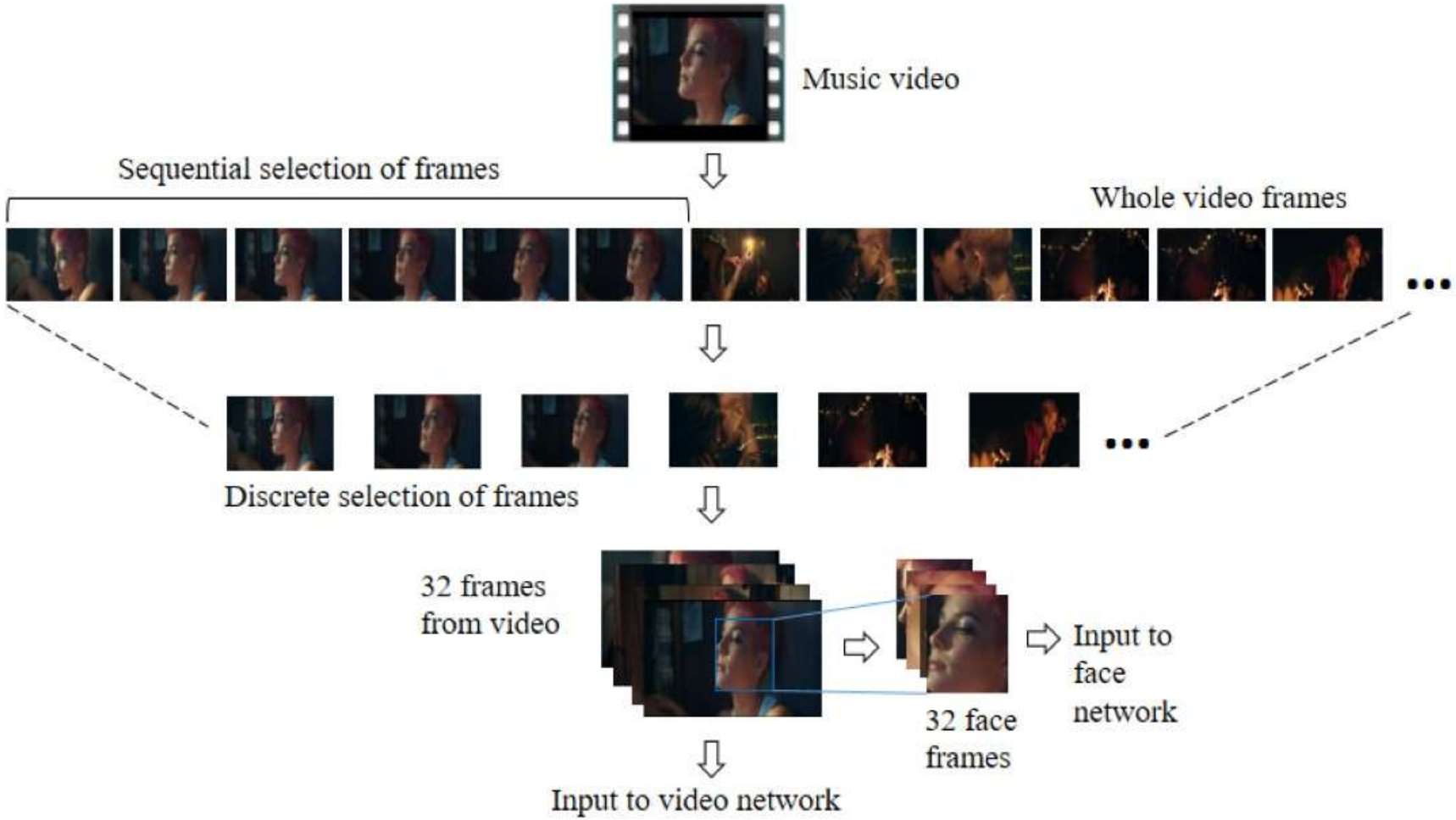
Video Frames

Video is a sequence of images

Extract frames from equal interval



Face3D: Sequential vs Discrete Frames



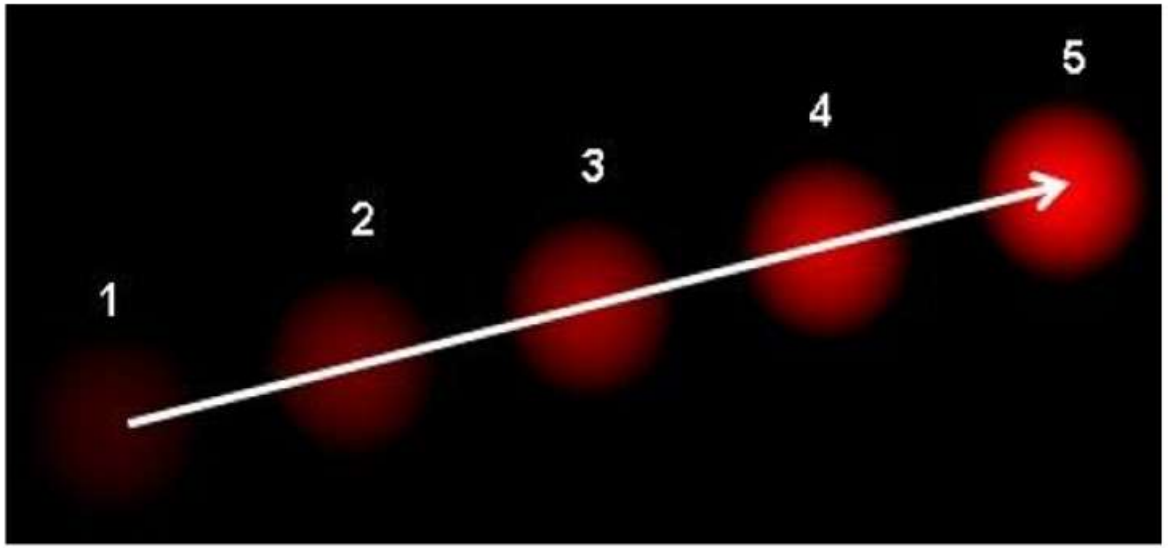
<< Let's Code it (Ecllipse) >>

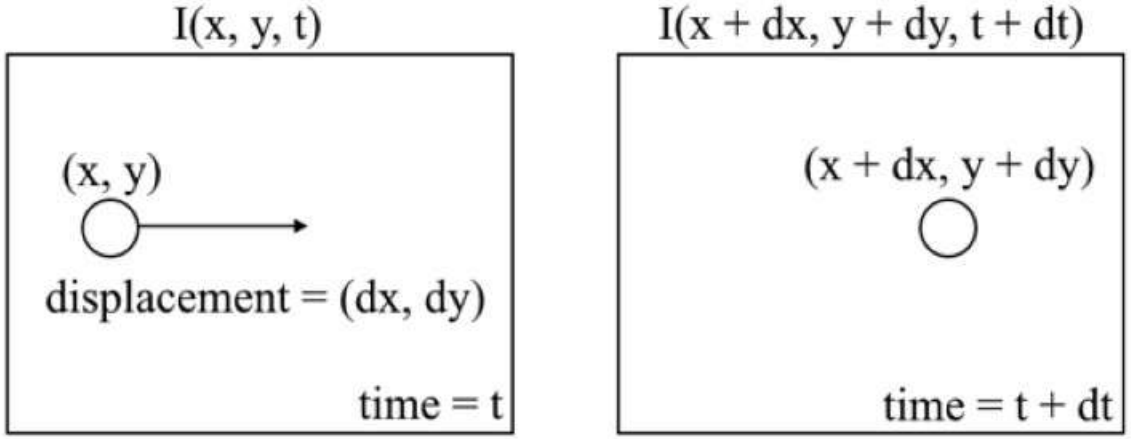
Optical Flow

What is optical flow?

Optical Flow is the motion of objects between consecutive frames of sequence.

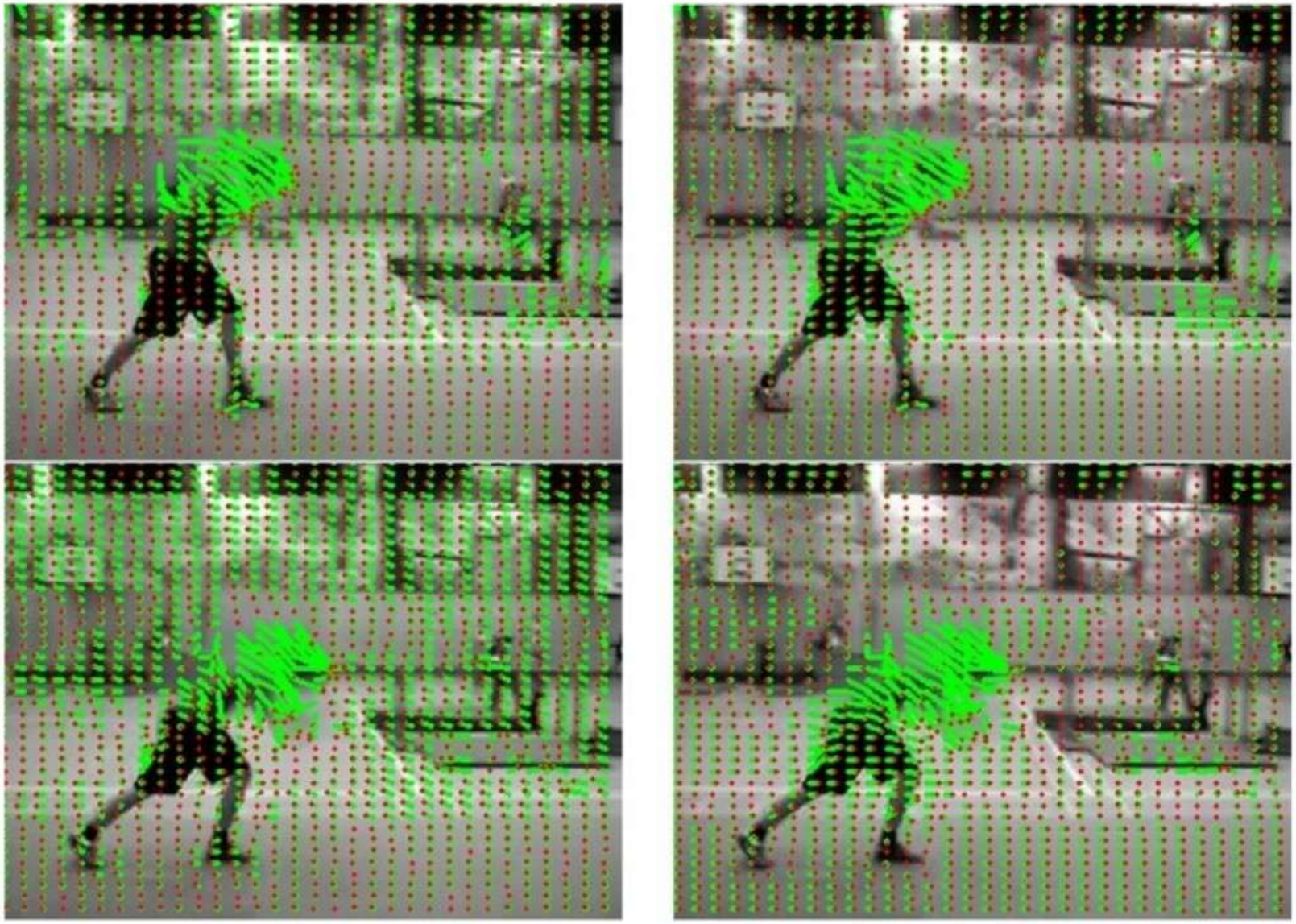
Either the camera is moving, or the scene is moving.





Optical Flow equation $u = \frac{dx}{dt}$; $v = \frac{dy}{dt}$

Example: Sports Analytics: Motion Analysis for Performance Improvement and Injury Prevention



Types

Sparse Optical Flow: Computes motion vectors only for a subset of pixels, usually those that are deemed to be more significant or easier to track (like corners or edges).

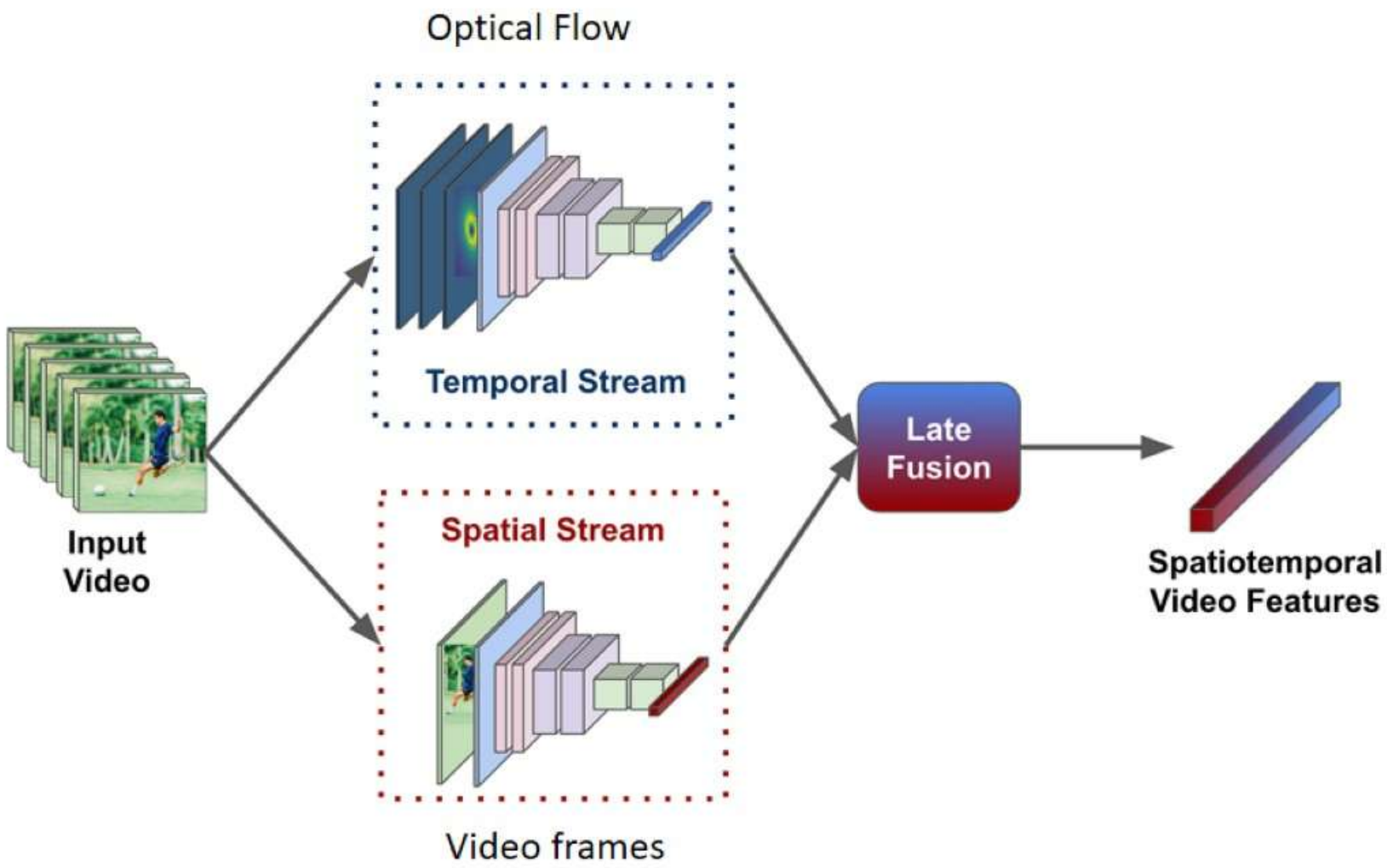
Dense Optical Flow: Computes motion vectors for every pixel in the frame.

(a) Sparse Optical Flow – Lukas Kanade

(b) Dense Optical Flow - Gunnar Farneback

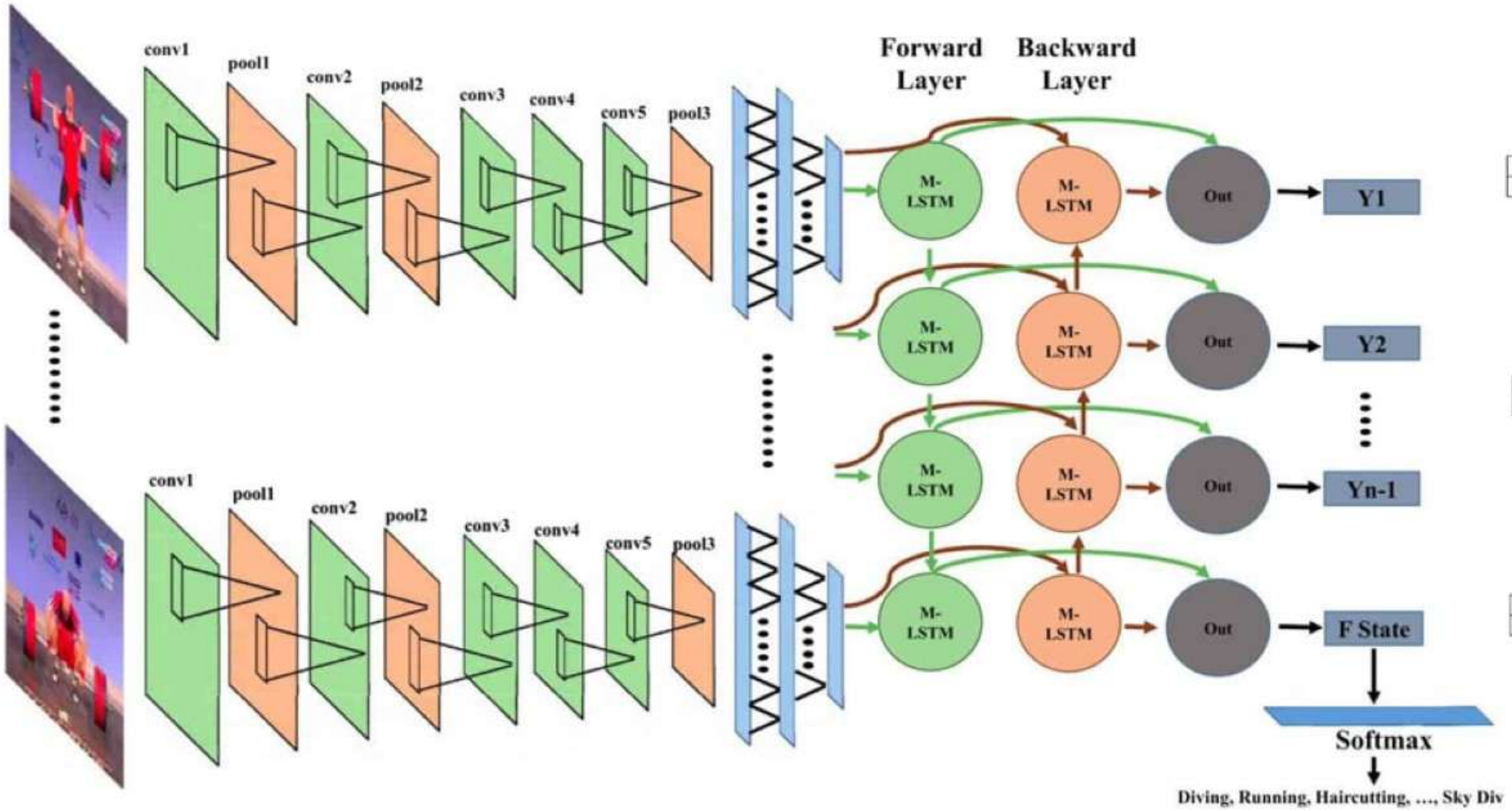
<< Let's Code it (1) >>

Video Processing: Assignment



Video Classification
CONVOLUTIONS

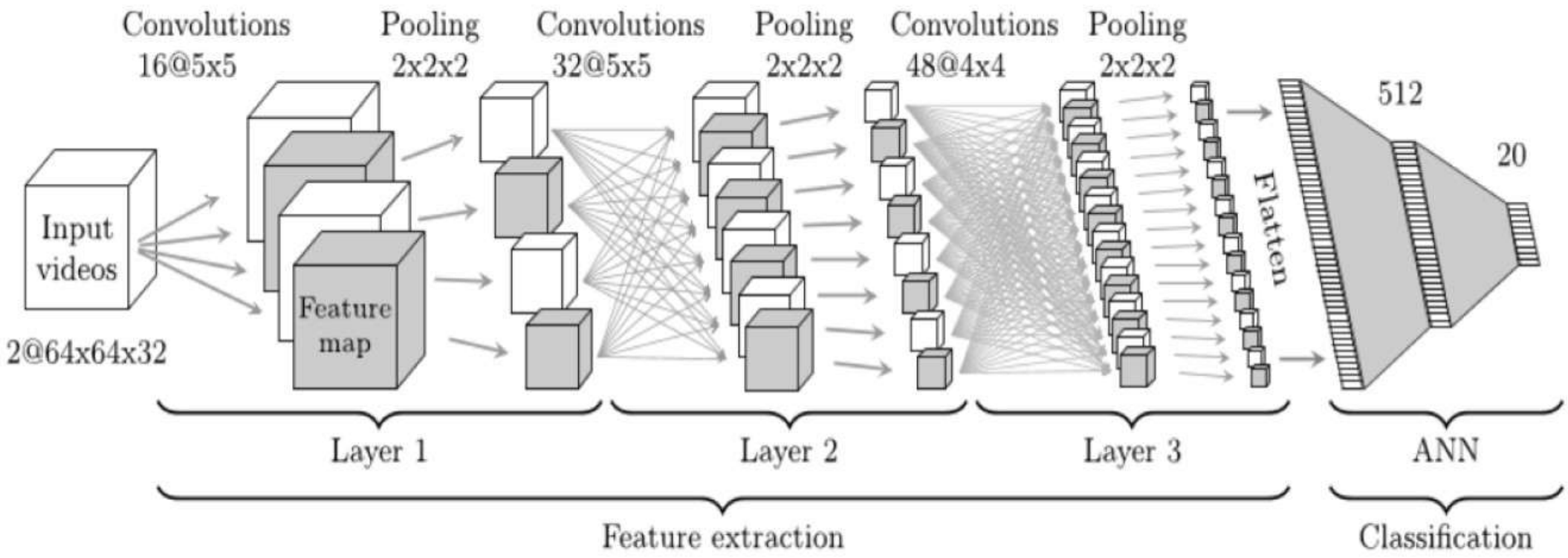
Video Processing: 2D Conv.



time_distributed_input	input:	[(None, 20, 64, 64, 3)]
InputLayer	output:	[(None, 20, 64, 64, 3)]
time_distributed(conv2d)	input:	(None, 20, 64, 64, 3)
TimeDistributed(Conv2D)	output:	(None, 20, 64, 64, 16)
time_distributed_1(max_pooling2d)	input:	(None, 20, 64, 64, 16)
TimeDistributed(MaxPooling2D)	output:	(None, 20, 16, 16, 16)
time_distributed_2(dropout)	input:	(None, 20, 16, 16, 16)
TimeDistributed(Dropout)	output:	(None, 20, 16, 16, 16)
time_distributed_3(conv2d_1)	input:	(None, 20, 16, 16, 16)
TimeDistributed(Conv2D)	output:	(None, 20, 16, 16, 32)
time_distributed_4(max_pooling2d_1)	input:	(None, 20, 16, 16, 32)
TimeDistributed(MaxPooling2D)	output:	(None, 20, 4, 4, 32)
time_distributed_5(dropout_1)	input:	(None, 20, 4, 4, 32)
TimeDistributed(Dropout)	output:	(None, 20, 4, 4, 32)
time_distributed_6(conv2d_2)	input:	(None, 20, 4, 4, 32)
TimeDistributed(Conv2D)	output:	(None, 20, 4, 4, 64)
time_distributed_7(max_pooling2d_2)	input:	(None, 20, 4, 4, 64)
TimeDistributed(MaxPooling2D)	output:	(None, 20, 2, 2, 64)
time_distributed_8(dropout_2)	input:	(None, 20, 2, 2, 64)
TimeDistributed(Dropout)	output:	(None, 20, 2, 2, 64)
time_distributed_9(conv2d_3)	input:	(None, 20, 2, 2, 64)
TimeDistributed(Conv2D)	output:	(None, 20, 2, 2, 64)
time_distributed_10(max_pooling2d_3)	input:	(None, 20, 2, 2, 64)
TimeDistributed(MaxPooling2D)	output:	(None, 20, 1, 1, 64)
time_distributed_11(flatten)	input:	(None, 20, 1, 1, 64)
TimeDistributed(Flatten)	output:	(None, 20, 64)
lstm	input:	(None, 20, 64)
LSTM	output:	(None, 32)
dense	input:	(None, 32)
Dense	output:	(None, 4)

<< Let's Code it (2) >>

3D CNN



https://github.com/harvitronix/five-video-classification-methods

C3D: Video Net

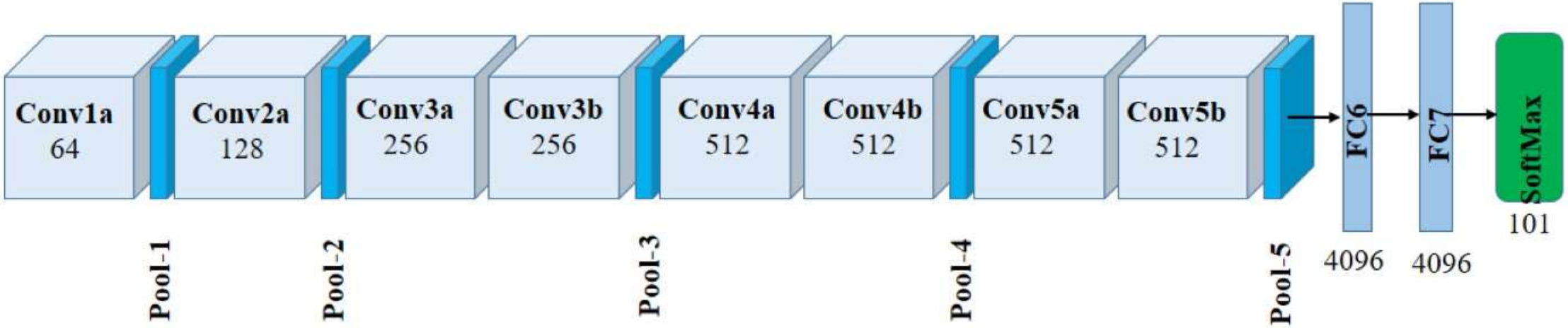
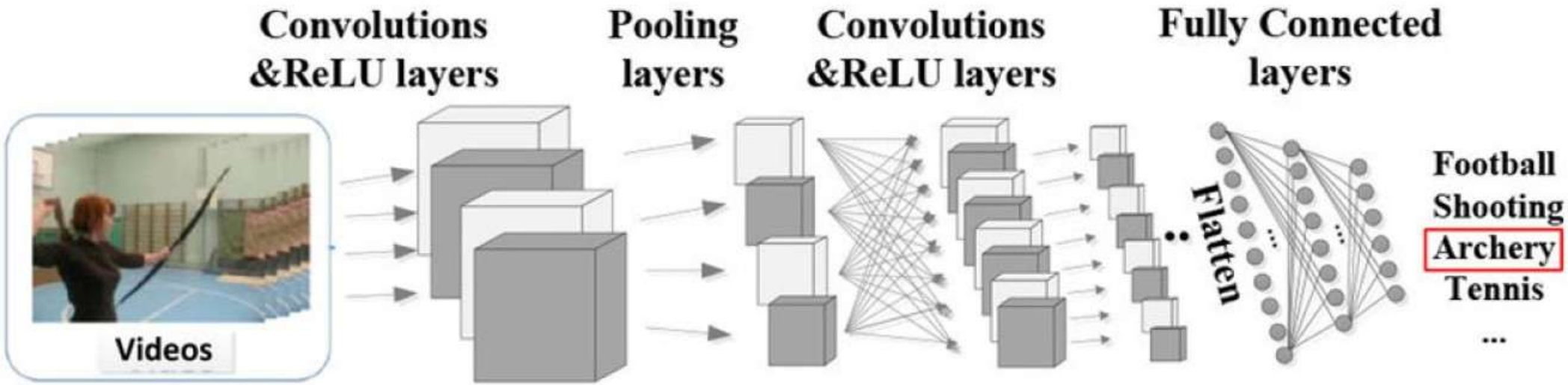


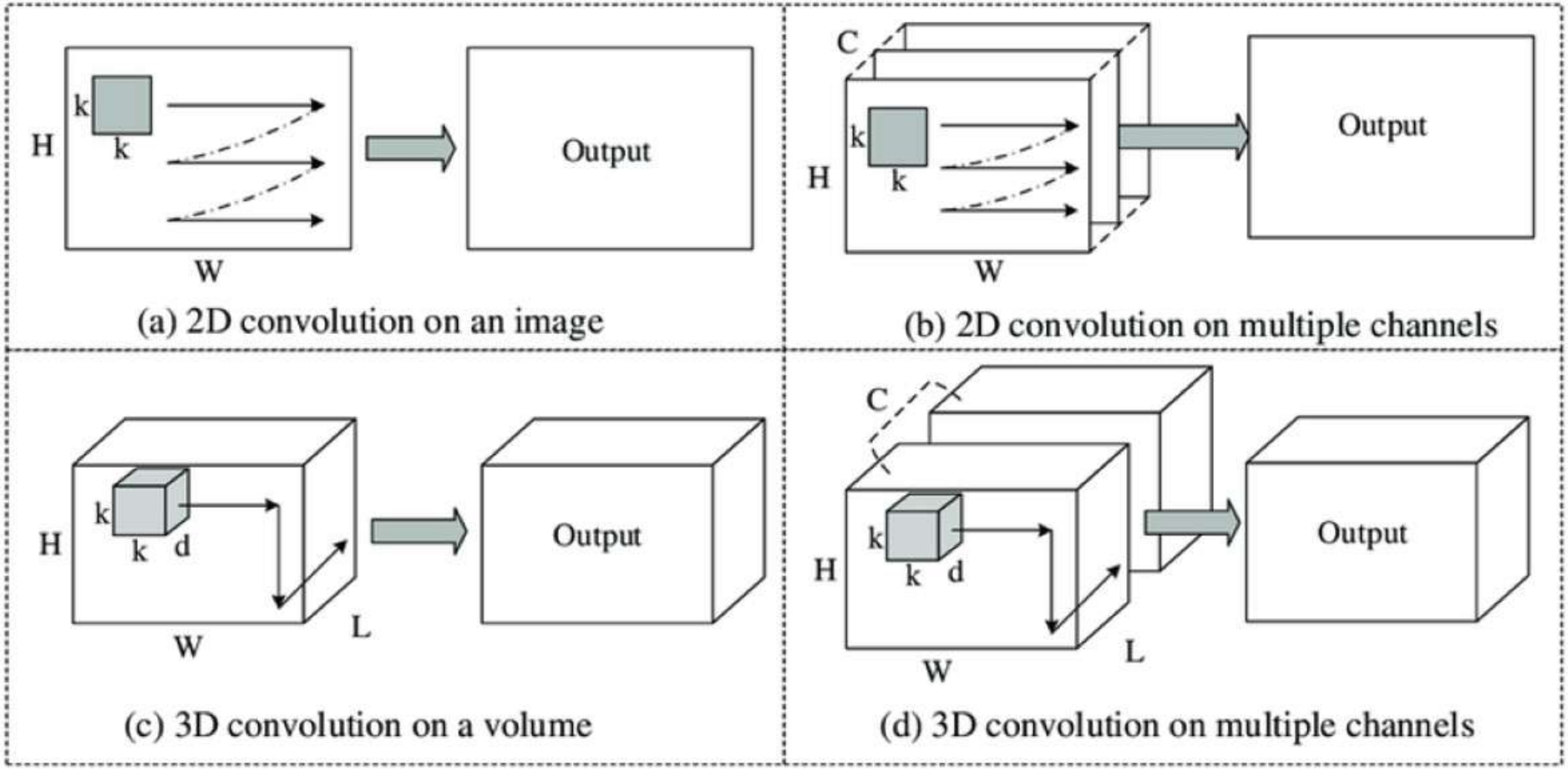
Fig. C3D Architecture

<< Let's Code it (3) >>

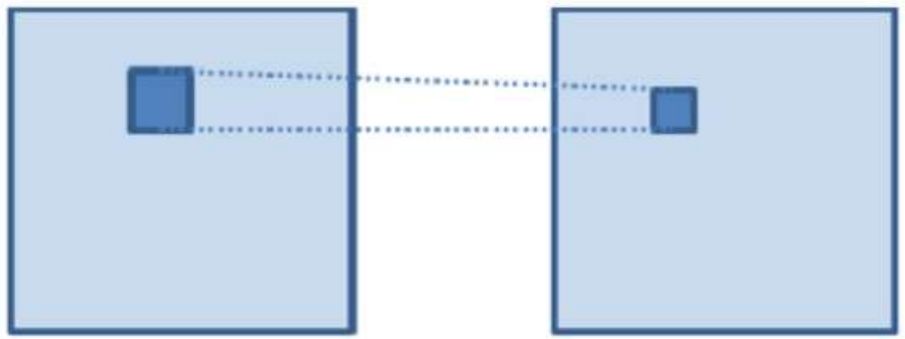
Video Classification: 3D Conv



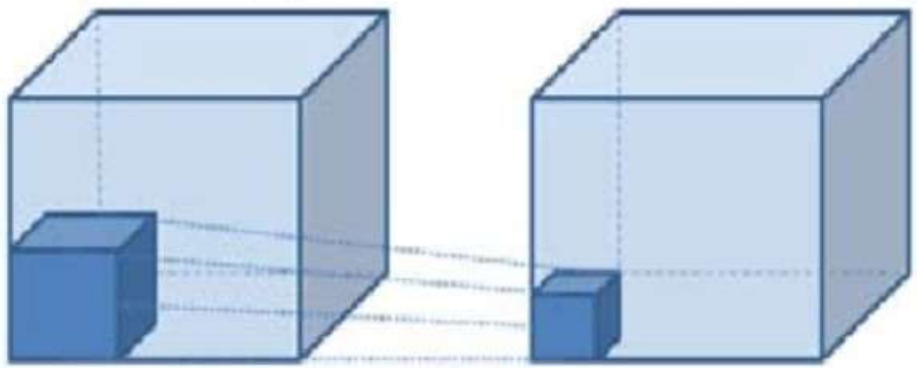
2D and 3D Convolution



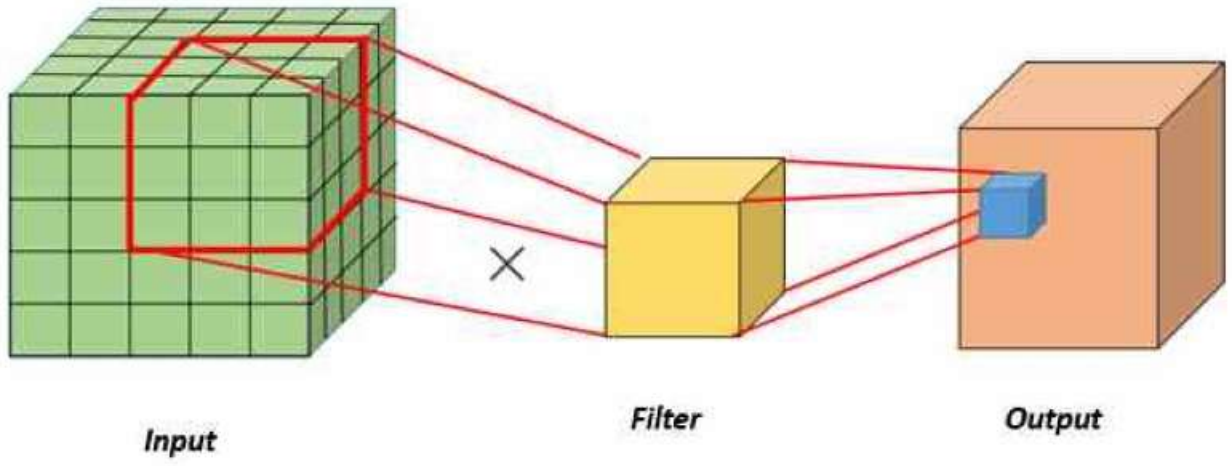
2D and 3D Convolution



(*a*) 2D convolution

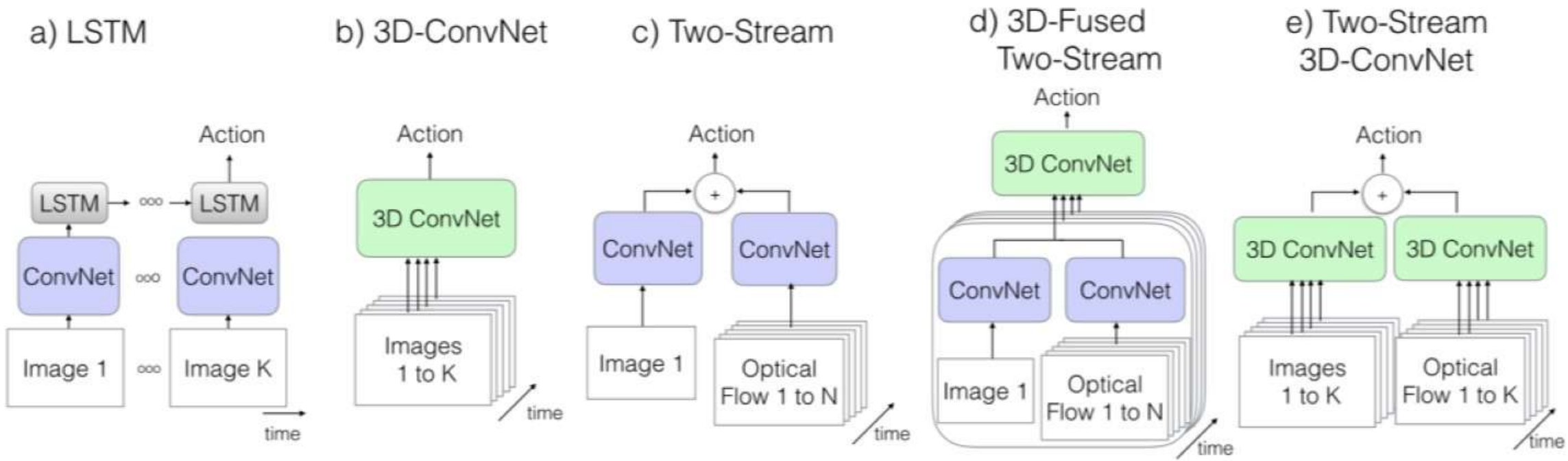


(*b*) 3D convolution



3D Convolution

Video Architecture Comparison



Method	#Params	Training		Testing	
		# Input Frames	Temporal Footprint	# Input Frames	Temporal Footprint
ConvNet+LSTM	9M	25 rgb	5s	50 rgb	10s
3D-ConvNet	79M	16 rgb	0.64s	240 rgb	9.6s
Two-Stream	12M	1 rgb, 10 flow	0.4s	25 rgb, 250 flow	10s
3D-Fused	39M	5 rgb, 50 flow	2s	25 rgb, 250 flow	10s
Two-Stream I3D	25M	64 rgb, 64 flow	2.56s	250 rgb, 250 flow	10s

Table 1. Number of parameters and temporal input sizes of the models.

Video classification

- Different 3D networks
  - Convolution 3D (C3D)
  - I3D

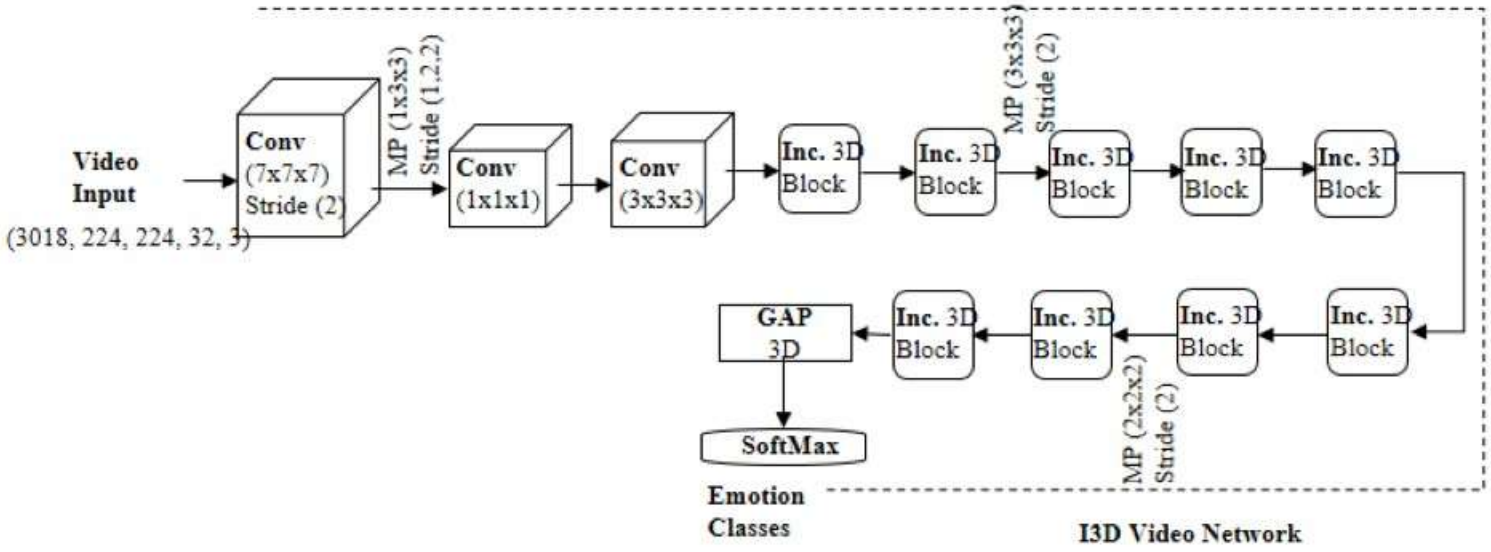


Fig. I3D network for video classification

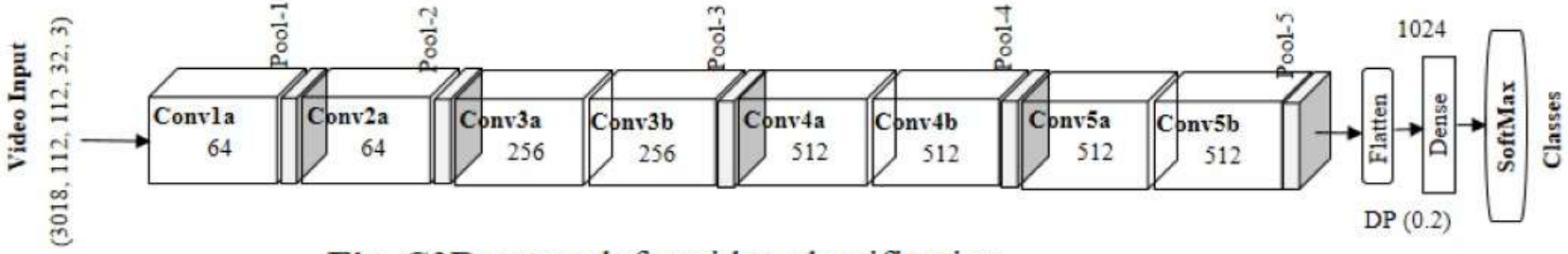
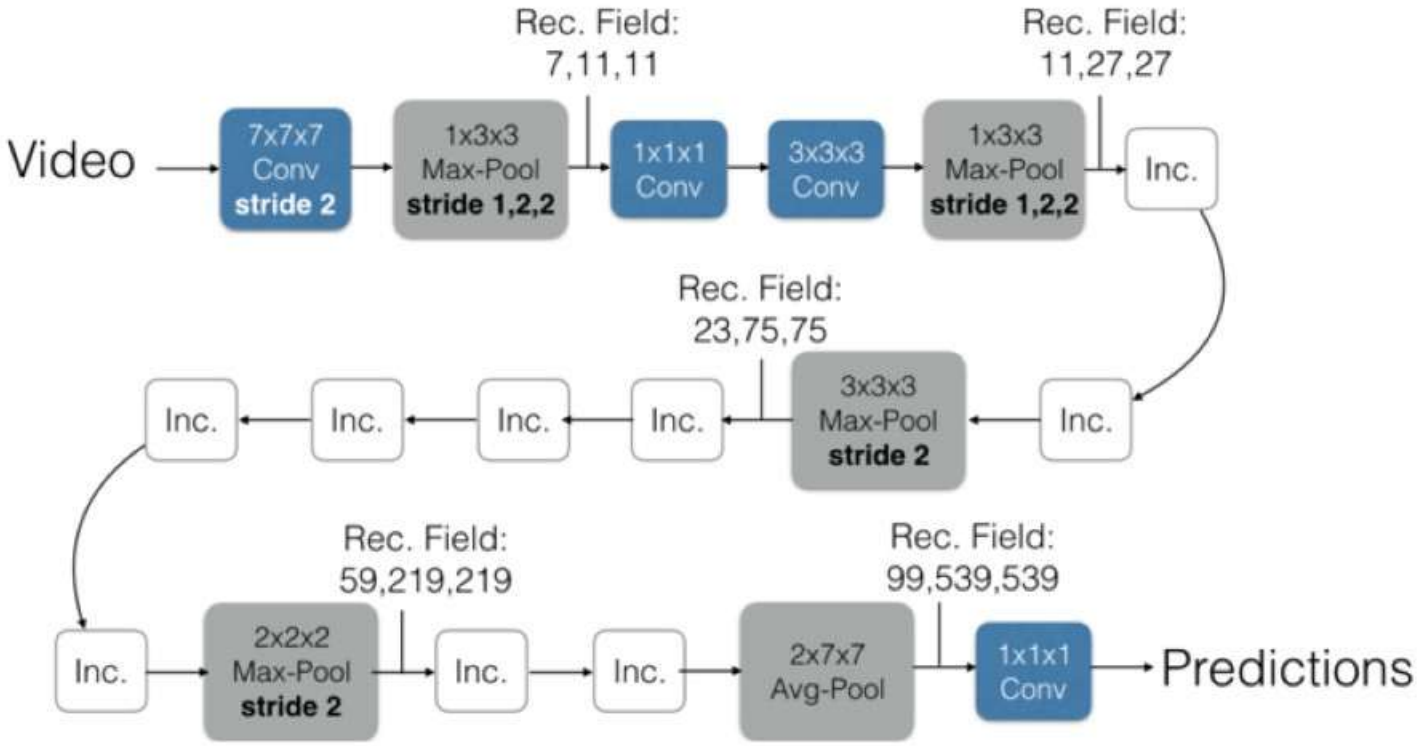


Fig. C3D network for video classification

I3D Architecture



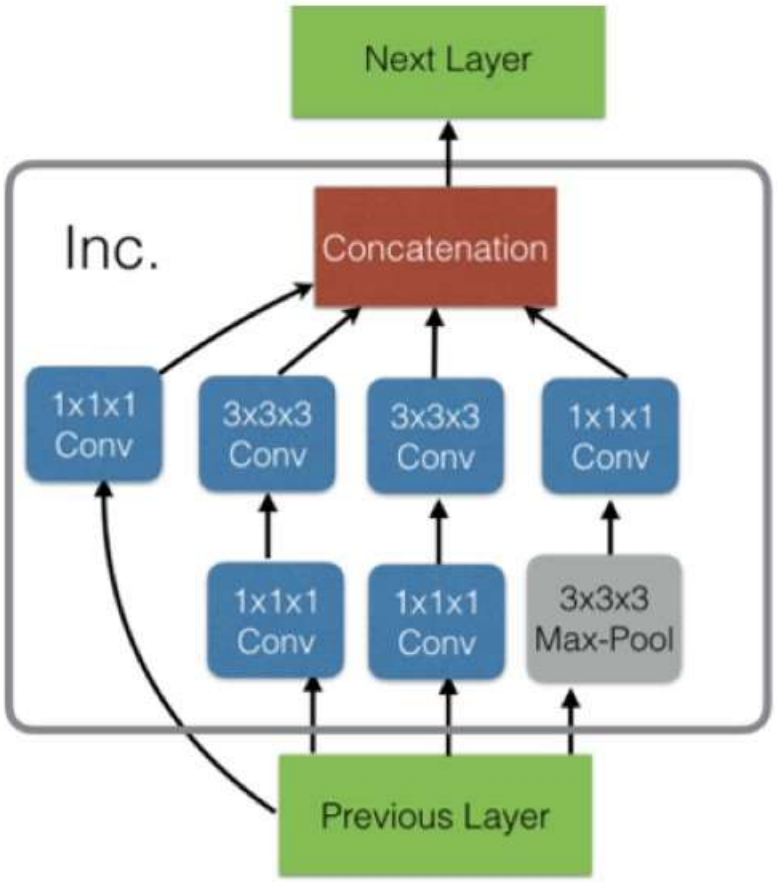
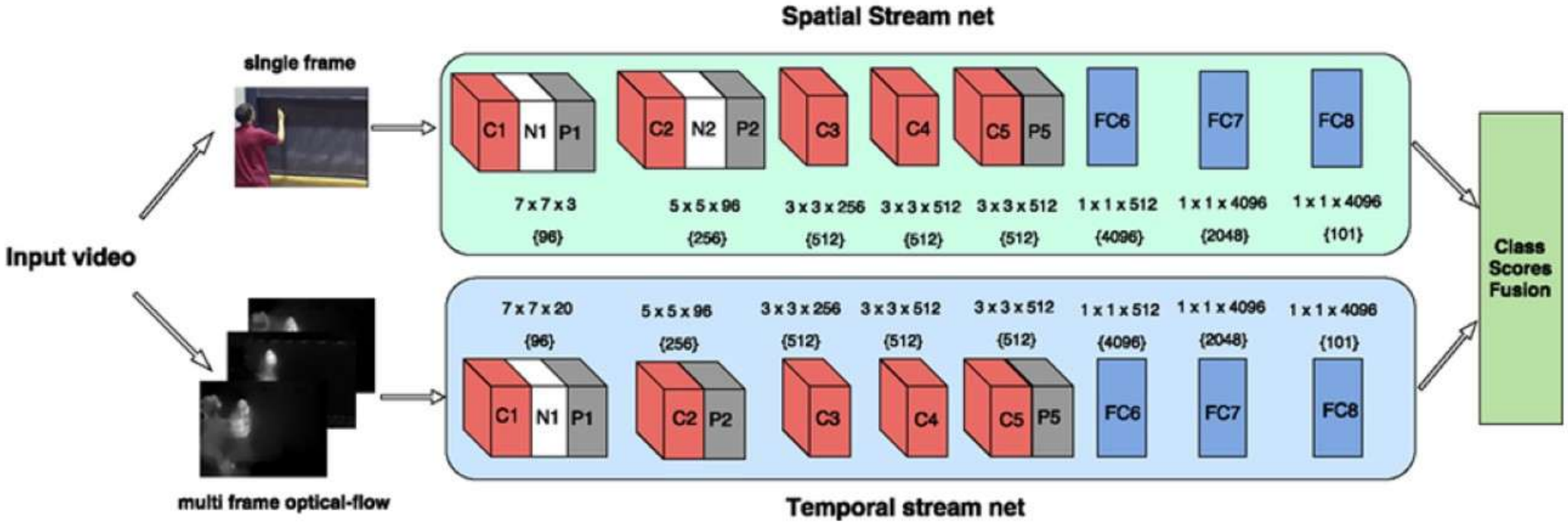
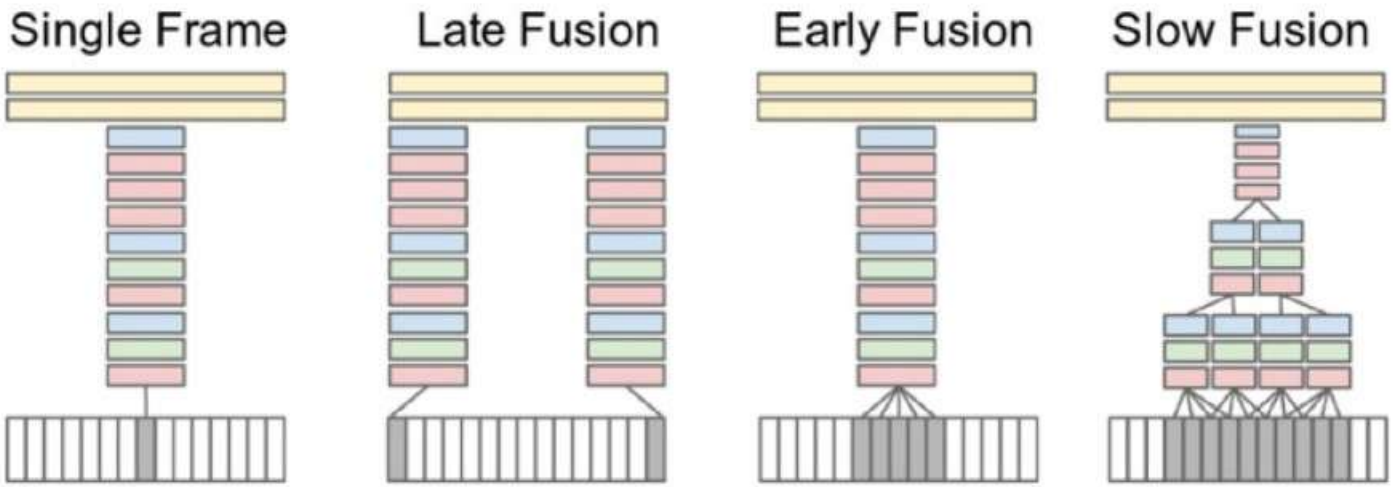


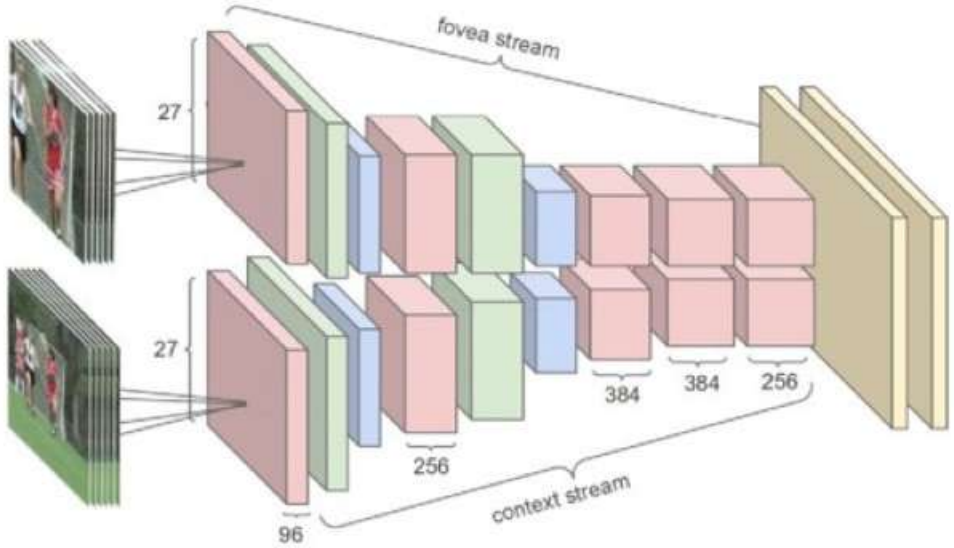
Figure. The Inflated Inception-V1 architecture (left) and its detailed inception submodule (right). The strides of convolution and pooling operators are 1 where not specified, and batch normalization layers, ReLu's and the softmax at the end are not shown.

Video Processing: Image + Optical Flow



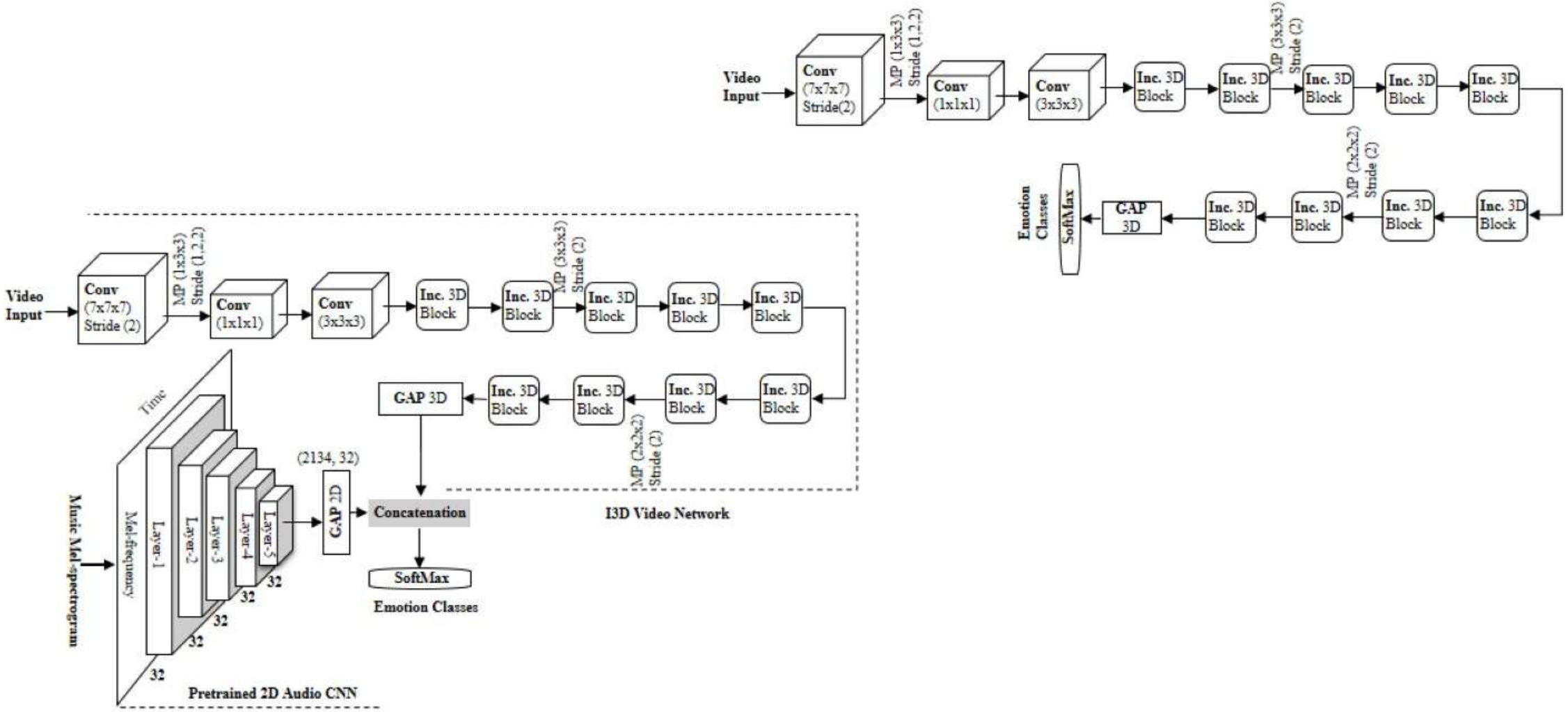
Information Fusion in Video





Large-scale Video Classification with Convolutional Neural Networks, CVPR 2014

Multimodal Architecture (I3D + Raw Audio 1DCNN)



Video Action Recognition

- Automatically identifying and categorizing human actions or activities from video data
- Tasks in Video Action Recognition:
  - **Classification:** The primary task involves classifying the actions depicted in a video into predefined categories. Examples include actions like walking, running, jumping, waving, or handshaking.
  - **Localization:** In some cases, VAR might not only identify the action but also localize it temporally (start and end time) or spatially (bounding box) within the video frame.
  - **Detection:** This advanced task aims to identify and localize all instances of specific actions occurring within a video, potentially involving multiple actions happening simultaneously.

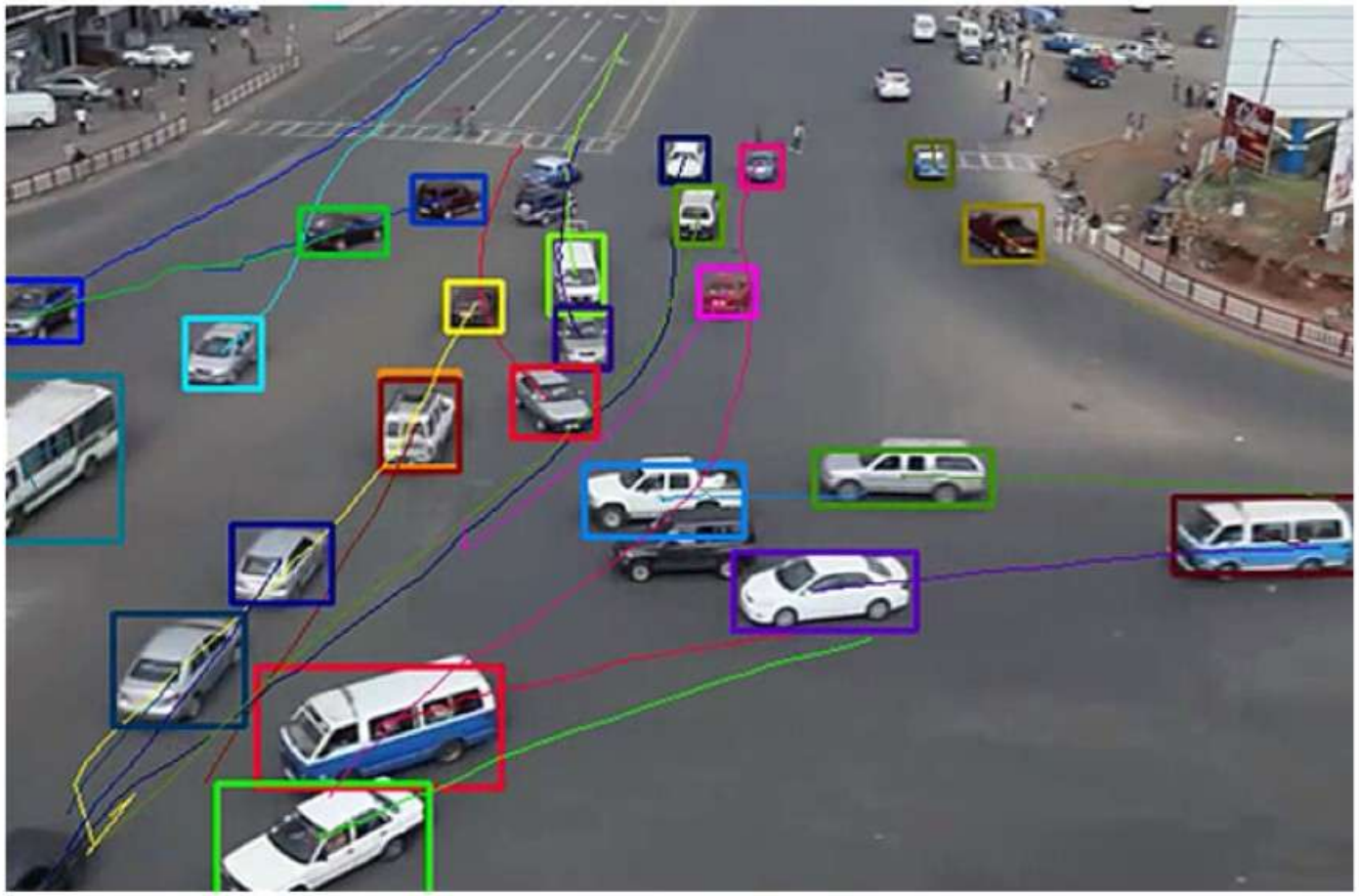
Object tracking and detection

- Object Detection:
  - **Goal:** Identify and locate objects within an image or video frame.
  - **Output:** Bounding boxes or keypoints around the detected objects along with class labels (e.g., car, person, bicycle).

- Object Tracking:
  - **Goal:** Follow the movement of a detected object across consecutive video frames.
  - **Output:** Trajectory or path of the object over time, potentially with additional information like bounding boxes in each frame.

Detection	Tracking
Detection focuses on identifying objects in a single frame.	Tracking follows their movement over time.
Detection provides object location and class.	Tracking adds the temporal element (trajectory) in the detection result

Video object detection



Video object tracking

Some solutions: https://github.com/gaomingqi/Track-Anything

	Video action recognition	Video object tracking
Concepts	Focuses on **identifying and classifying the actions** depicted in a video. It aims to understand **what** is happening in the video (e.g., walking, jumping, waving).	Focuses on **following the movement of objects** within a video. It aims to understand **where** objects are located and how they move over time.
Output	Outputs labels for the actions happening in the video (e.g., "running," "handshaking") or probabilities of different actions occurring. It might also include temporal information like the start and end times of actions.	Outputs the trajectories or paths of objects as they move through the video. This can be represented by bounding boxes around the object in each frame or more detailed information like position and velocity.
Methods	Often relies on deep learning models, particularly Convolutional Neural Networks (CNNs) or Recurrent Neural Networks (RNNs) to analyze video frames and sequences.	Employs various techniques like correlation-based tracking, Kalman filters, or even leverages detection outputs from object detection algorithms.
Use	Used in video surveillance (anomaly detection), human-computer interaction (gesture recognition), sports analytics (analyzing player actions), and video indexing/retrieval (searching videos based on actions).	Valuable for self-driving cars (tracking pedestrians and vehicles), traffic monitoring (analyzing traffic flow), video surveillance (tracking objects of interest), and human-computer interaction (tracking user movements).

Assignments

1. Video classification using 2D convolution
2. Video classification using 3D convolution
3. Multimodal information fusion (Audio + Video)
4. Optical Flow + Convolution for video classification
5. I3D Video Classification
6. Attention in Video Classification
7. Video classification with various feature fusion (Late, early, slow)
8. Video action recognition
9. Video object detection
10. Video tracking
11. Video Segmentation
12. Video processing using transformer architecture
13. Face recognition in video data

Thank You

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